

Executive Summary

NHS Health Economic Forecasting Dashboard

Data Summary

Eight of eleven NHS performance series produce forecasts reliable enough to underwrite investment decisions at 1–4 quarter horizons; the remaining three (GP appointments) are excluded due to insufficient history. Two findings follow directly from this analysis.

- Sustained demand for NHS capacity-relief models (private elective capacity, insourcing, diagnostics, step-down care), supported by clear post-COVID structural shifts across core capacity series.
- Caution is warranted for any thesis premised on durable NHS understaffing. Workforce metrics show a clear deviation from pre-pandemic trend, running counter to a "workforce is falling behind" narrative.

Forecast reliability by series

Tier	Series	Application
Decision-grade	RTT waiting list, A&E attendances, Workforce FTE, Nurse FTE, Doctor FTE, Bed occupancy, RTT% (18wk), A&E 12h breach*	Point estimates at h=1,4 and directional trend only at h=8,12
Excluded	GP appointments (3 series)	Insufficient history (only 11 quarters per series)

*This series shows the cleanest forecast diagnostics of the eight but is based on a shorter history than the other series. Point accuracy is strong; however, binding SLA or contract terms should not be anchored to this series specifically until additional history is available. This reflects forecast accuracy rather than a COVID-era structural shift. For instance, RTT% shows a clean, well-calibrated forecast but no significant break effect (see Investment Implications).

Investment Implications

Capacity pressure (high confidence for bed occupancy, directionally supported elsewhere): Bed occupancy has dropped decisively below its pre-pandemic trend (approximately 23% lower, equivalent to roughly 34,700 fewer beds than the historical trajectory would predict). This is a well-supported, precisely estimated finding. The RTT waiting list shows a similarly clear and durable shift, consistent with sustained pressure on elective and diagnostic capacity; a single headline figure for its magnitude is not presented here given sensitivity to modelling assumptions (see technical writeup). A&E attendances' forecast is calibrated for accuracy rather than for detecting a structural shift and should therefore be read as supporting context for the capacity-pressure thesis rather than direct evidence of one.

Workforce (moderate confidence, runs counter to the prevailing understaffing narrative): NHS staffing has consistently grown faster than a trend-based model would predict, not slower, across all three workforce series (Doctors, Nurses, Total FTE). Any thesis premised on structural NHS understaffing as a durable opening for private-sector alternatives should treat this as a material data point requiring further consideration. This pattern is consistent with a gradual institutional shift rather than a one-off policy adjustment, and is therefore less likely to reverse quickly.

Scope: These implications are drawn directly from the dashboard's outputs. They constitute a data-backed starting point for market sizing, competitive analysis, and policy diligence, and are not a substitute for such work.

Methodological Assurance (For Technical Review)

- Every series was validated through rolling-origin backtesting. For bed occupancy, reported confidence intervals were formally confirmed against actual historical accuracy via a Kupiec unconditional coverage test, which passed at every tested horizon ($p > 0.05$). This formal coverage confirmation has not yet been established for every series, notably A&E 12h breach (only 22 historical observations) and RTT (interval calibration still being finalised; see technical writeup).
- The bed occupancy structural shift is precisely estimated: coefficient $-34,680$ beds, robust SE $1,520$, $p < 0.0001$. The RTT waiting list shows a similarly significant shift ($p < 0.0001$), though its magnitude is sensitive to long-run trend extrapolation assumptions and is deliberately not reported as a single point figure. Both are quantified via interrupted time-series regression with heteroskedasticity-robust standard errors, rather than a provisional adjustment.
- A&E attendances' production model includes no structural break term and therefore carries no p-value or coefficient for the COVID-era shift. However, its inclusion above is qualitative only.
- Any model configuration diverging from statistical best practice was logged and escalated, with production specifications adopted only against documented justification. This includes formal testing of three competing configurations for the two weakest-performing series (Nurse and Doctor FTE).

Full model specifications, coefficients, per-series diagnostics, and documented limitations regarding autocorrelation-robust (HAC) standard errors are available in the accompanying technical writeup.